Statistical Machine Learning - HW 4 (Solutions)

Homework 4 Problem

For leave-one-out cross validation (or equivalently c-fold cross validation with $c = n$) the cross validation error E_{val} for ridge regression actually has a closed-form solution

$$E_{\text{val}} = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{\hat{y}_i - y_i}{1 - [H(\gamma)]_{ii}} \right)^2, \tag{1}$$

where $\hat{y}_i$ is the prediction of y_i when the model is learned from all n data points (no data point is left out), the regularization parameter γ and $[H(\gamma)]_{ii}$ is element (i, i) of the matrix

$$H(\gamma) = X(X^\top X + \gamma I)^{-1} X^\top.$$

In this problem, we will let $x_i^\top$ denote the entire ith row of X.

(a)

Let X_{-i} denote the matrix X where row i is removed, and y_{-i} is the column vector y with element i removed.

1. Show that $X_{-i}^\top X_{-i} = X^\top X - x_i x_i^\top$

Write the Gram matrix as a sum over rows:

$$X^\top X = \sum_{j=1}^{n} x_j x_j^\top .$$

Removing row i gives

$$X_{-i}^\top X_{-i} = \sum_{\substack{j=1 \\ j\neq i}}^{n} x_j x_j^\top = \sum_{j=1}^{n} x_j x_j^\top - x_i x_i^\top = X^\top X - x_i x_i^\top .$$

2. Show that $X_{-i}^\top y_{-i} = X^\top y - x_i y_i$

Similarly,

$$X^\top y = \sum_{j=1}^{n} x_j y_j .$$

Therefore, $$ X_{-i}^ y_{-i} = *{*\substack{j=1\\ j\neq i}}^n x_j y_j = j=1^n x_j y_j - x_i y_i = X^ y - x_i y_i.

3. Show that $[H(\gamma)]_{ii} = x_i^\top (X^\top X + \gamma I)^{-1} x_i$

Let

$$A := (X^\top X + \gamma I)^{-1}.$$

Then

$$H(\gamma) = XAX^\top .$$

The (i, i) entry of $XAX^\top$ is the ith row of X times A times the ith column of $X^\top$:

$$[H(\gamma)]_{ii} = x_i^\top A x_i = x_i^\top (X^\top X + \gamma I)^{-1} x_i .$$

(b)

We use ridge regression in closed form. Define

$$A := X^\top X + \gamma I, \qquad \hat\beta := A^{-1}X^\top y.$$

For the leave-i-out fit,

$$\hat\beta_{-i} = (X_{-i}^\top X_{-i} + \gamma I)^{-1} X_{-i}^\top y_{-i}.$$

Using part (a),

$$X_{-i}^\top X_{-i} + \gamma I = (X^\top X - x_i x_i^\top) + \gamma I = A - x_i x_i^\top,$$

and

$$X_{-i}^\top y_{-i} = X^\top y - x_i y_i.$$

Hence

$$\hat\beta_{-i} = (A - x_i x_i^\top)^{-1}(X^\top y - x_i y_i).$$

Now apply the matrix inversion lemma (with $v = x_i$):

$$(A - x_i x_i^\top)^{-1} = A^{-1} + \frac{A^{-1}x_i x_i^\top A^{-1}}{1 - x_i^\top A^{-1} x_i}.$$

Let

$$h_{ii} := x_i^\top A^{-1} x_i = [H(\gamma)]_{ii}.$$

Then

$$\hat\beta_{-i} = \left(A^{-1} + \frac{A^{-1}x_i x_i^\top A^{-1}}{1 - h_{ii}}\right)(X^\top y - x_i y_i).$$

Distribute:

$$\hat\beta_{-i} = A^{-1}X^\top y - A^{-1}x_i y_i + \frac{A^{-1}x_i x_i^\top A^{-1}X^\top y}{1 - h_{ii}} - \frac{A^{-1}x_i x_i^\top A^{-1}x_i\, y_i}{1 - h_{ii}}.$$

Recognize $A^{-1}X^\top y = \hat\beta$, and

$$x_i^\top A^{-1}X^\top y = x_i^\top \hat\beta = \hat y_i, \qquad x_i^\top A^{-1} x_i = h_{ii}.$$

So

$$\hat\beta_{-i} = \hat\beta - A^{-1}x_i y_i + \frac{A^{-1}x_i \hat y_i}{1 - h_{ii}} - \frac{A^{-1}x_i h_{ii} y_i}{1 - h_{ii}}.$$

Combine the terms involving $A^{-1}x_i$ by putting $\hat\beta - A^{-1}x_i y_i$ over the common denominator:

$$\hat\beta - A^{-1}x_i y_i = \hat\beta + \frac{-A^{-1}x_i y_i(1 - h_{ii})}{1 - h_{ii}}.$$

Thus

$$\hat{\beta}_{-i} = \hat{\beta} + \frac{A^{-1}x_i\left[\hat{y}_i - h_{ii}y_i - y_i(1-h_{ii})\right]}{1-h_{ii}} = \hat{\beta} + \frac{A^{-1}x_i(\hat{y}_i - y_i)}{1-h_{ii}}.$$

Since $A^{-1} = (X^\top X + \gamma I)^{-1}$ and $h_{ii} = [H(\gamma)]_{ii}$, we obtain

$$\boxed{\hat{\beta}_{-i} = \hat{\beta} + \frac{1}{1-[H(\gamma)]_{ii}}(X^\top X + \gamma I)^{-1}x_i(\hat{y}_i - y_i).}$$

(c)

Start from

$$E_{\text{val}} = \frac{1}{n}\sum_{i=1}^{n}(\hat{\beta}_{-i}^\top x_i - y_i)^2.$$

Using the expression from (b),

$$\hat{\beta}_{-i}^\top x_i = \left(\hat{\beta} + \frac{1}{1-h_{ii}}A^{-1}x_i(\hat{y}_i - y_i)\right)^\top x_i = \hat{\beta}^\top x_i + \frac{1}{1-h_{ii}}(\hat{y}_i - y_i)\, x_i^\top A^{-1}x_i.$$

But $\hat{\beta}^\top x_i = \hat{y}_i$ and $x_i^\top A^{-1}x_i = h_{ii}$, so

$$\hat{\beta}_{-i}^\top x_i = \hat{y}_i + \frac{h_{ii}}{1-h_{ii}}(\hat{y}_i - y_i).$$

Therefore the leave-one-out residual is

$$\hat{\beta}_{-i}^\top x_i - y_i = (\hat{y}_i - y_i) + \frac{h_{ii}}{1-h_{ii}}(\hat{y}_i - y_i) = \frac{1}{1-h_{ii}}(\hat{y}_i - y_i).$$

Plug into the CV error:

$$E_{\text{val}} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{\hat{y}_i - y_i}{1-h_{ii}}\right)^2 = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{\hat{y}_i - y_i}{1-[H(\gamma)]_{ii}}\right)^2,$$

which is exactly eq. (1).

(d)

Eq. (1) lets you compute the leave-one-out cross validation error for ridge regression **without refitting the model** n times. In practice, you fit ridge once for a given γ, compute the fitted values $\hat{y}$ and the diagonal of the hat matrix $H(\gamma)$, and then get the LOOCV error by the simple formula. This makes it efficient to **select** γ (e.g., by searching over a grid of γ values) even when n is large.